

Connecting Amazon RDS to an EC2 Instance Using MySQL Client

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon RDS MySQL** database was deployed inside a VPC and connected securely to a Linux-based **Amazon EC2** instance acting as a MySQL client. Separate security groups were created for the EC2 instance and RDS database to follow a more realistic production-style architecture. Public access was disabled on the database, ensuring that connections could only originate from resources inside the VPC. After both resources were launched, the MySQL client was installed on the EC2 instance, the RDS endpoint was used to establish a connection, and sample database objects were created and verified. This demonstrated how application servers typically communicate with backend databases in real-world AWS environments.

Understanding Amazon RDS Private Database Access

Amazon RDS databases are typically deployed in private networks so they are not directly accessible from the internet. Instead, application servers or management servers running on **Amazon EC2** connect to the database using internal VPC networking. This improves security because database endpoints are hidden from public access while still allowing authorized applications to read and write data. Security Groups act as virtual firewalls, controlling which resources can communicate with the database.

Example:

An application server running on EC2 hosts an employee management portal. Employees submit information through the web application, and the EC2 server stores that data in an RDS MySQL database. Users never access the database directly; all communication flows through the application server, reducing security risks and exposing only the application interface to the internet.

Lab Steps

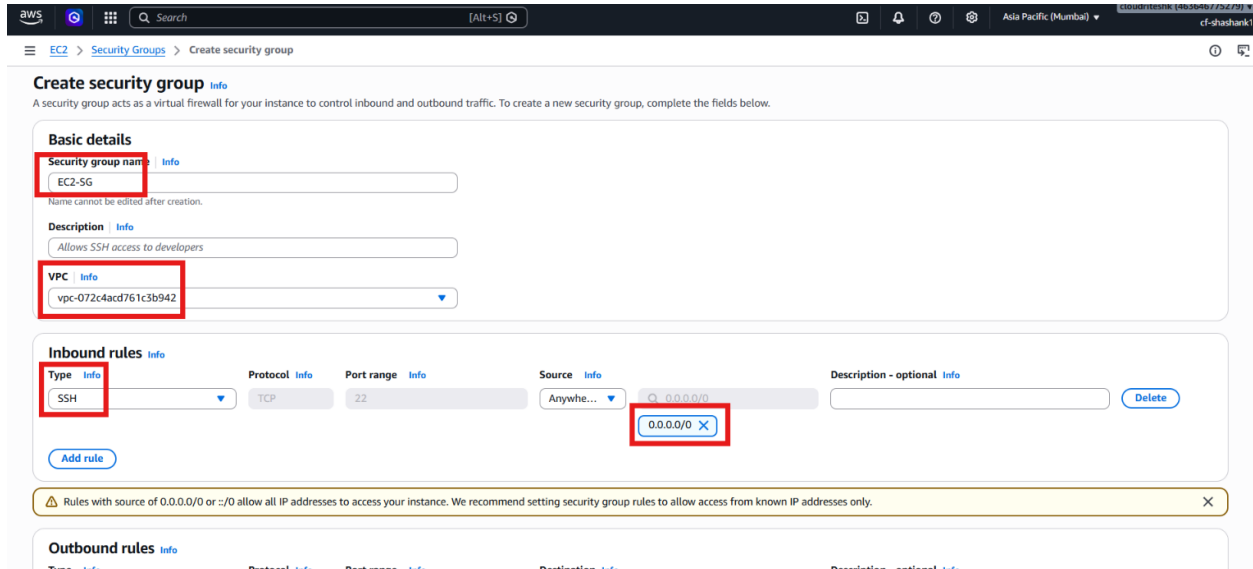
Step 1 – Create Security Group for EC2

- Sign in to the AWS Management Console.
- Navigate to **Amazon EC2** → **Security Groups**.
- Click **Create Security Group**.
- Enter the security group name **EC2-SG**.

- Add a description for the EC2 MySQL client instance.
- Select the default VPC.
- Configure the inbound rule:

Type	Protocol	Port
SSH	TCP	22

- Leave outbound rules as default (**All Traffic**).



Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)
EC2-SG
Name cannot be edited after creation.

Description [Info](#)
Allows SSH access to developers

VPC [Info](#)
vpc-072c4acd761c3b942

Inbound rules [Info](#)

Type [Info](#): SSH
Protocol [Info](#): TCP
Port range [Info](#): 22
Source [Info](#): Anywhere...
Destination [Info](#): 0.0.0.0/0
Description - optional [Info](#):
[Delete](#)

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type [Info](#):
Protocol [Info](#):
Port range [Info](#):
Destination [Info](#):
Description - optional [Info](#):

- Click **Create Security Group**.

Step 2 – Create Security Group for RDS

- Navigate to **Amazon EC2** → **Security Groups**.
- Click **Create Security Group**.
- Enter the security group name **RDS-SG**.
- Add a description for the RDS database.
- Select the same VPC used by the EC2 instance.
- Configure the inbound rule:

Type	Protocol	Port
MySQL/Aurora	TCP	3306

- Leave outbound rules as default (**All Traffic**).

- Click **Create Security Group**.

Step 3 – Create the Amazon RDS Database

- Navigate to **Amazon RDS** → **Databases**.
- Click **Create Database**.
- Select **Standard Create**.
- Choose **MySQL**.
- Select the default MySQL 8.x version.
- Choose **Free Tier**.
- Enter a database identifier such as **myappdb**.
- Configure master credentials:
 - Username: **admin**
 - Password: Your chosen password
- Keep the default instance configuration.
- Set storage to 20 GB.
- Select the default VPC.
- Disable **Public Access**.
- Under **VPC Security Groups**, remove the default security group.
- Attach **RDS-SG**.

Public access [Info](#)

☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
Choose existing VPC security groups

☐ Create new
Create new VPC security group

Existing VPC security groups

Choose one or more options

RDS-SG X

Availability Zone [Info](#)

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ Create an RDS Proxy [Info](#)
RDS Proxy automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061

If you don't select a certificate authority, RDS chooses one for you.

- Leave the remaining settings as default.
- Click **Create Database**.
- Wait until the database status changes to **Available**.

Step 4 – Launch the EC2 MySQL Client Instance

- Navigate to **Amazon EC2** → **Instances**.
- Click **Launch Instance**.
- Enter the name **MySQL-Client**.
- Select **Amazon Linux 2023**.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
MySQL-Client [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents My AMIs **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Linux 2023 kernel-6.1 AMI
ami-0c38355da6f6ba2b9 (64-bit (x86), uefi-preferred) / ami-07c687f88f7820e6 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...[read more](#)
ami-0c38355da6f6ba2b9

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as

[Cancel](#) [Launch instance](#) [Preview code](#)

- Choose **t3.micro**.
- Select an existing key pair.
- Ensure the instance is launched in the same VPC as the RDS instance.
- Enable a public IP address.

- Under **Network Settings**, select **Existing Security Group**.
- Attach **EC2-SG**.
- Click **Launch Instance**.

- Wait until the instance enters the **Running** state.

Step 5 – Connect EC2 to RDS

- Connect to the EC2 instance using SSH.

```
ssh -i your-key.pem ec2-user@EC2-Public-IP
```

- Update the package repository.

```
sudo dnf update -y
```



```
ec2-user@ip-172-31-6-40:~  
(3/5): mariadb105-10.5.29-1.amzn2023.0.1.x86_64.rpm 27 MB/s | 1.5 MB 00:00  
(4/5): mariadb105-common-10.5.29-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 28 kB 00:00  
(5/5): perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64.rpm 604 kB/s | 17 kB 00:00  
-----  
Total 13 MB/s | 1.8 MB 00:00  
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing      : 1/1  
  Installing     : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch 1/5  
  Installing     : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64 2/5  
  Installing     : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64 3/5  
  Installing     : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64 4/5  
  Installing     : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 5/5  
  Running scriptlet: mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 5/5  
  Verifying      : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64 1/5  
  Verifying      : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch 2/5  
  Verifying      : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 3/5  
  Verifying      : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64 4/5  
  Verifying      : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64 5/5  
  
Installed:  
mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64 mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch  
mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64  
perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64  
  
Complete!  
[ec2-user@ip-172-31-6-40 ~]$ mysql -h myapppdb.cypfi5rmsrdv.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
```

- Enter the database password when prompted.
- Verify that the MySQL prompt appears successfully.

```
ec2-user@ip-172-31-6-40:~  
  Preparing      : 1/1  
  Installing     : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch 1/5  
  Installing     : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64 2/5  
  Installing     : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64 3/5  
  Installing     : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64 4/5  
  Installing     : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 5/5  
  Running scriptlet: mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 5/5  
  Verifying      : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64 1/5  
  Verifying      : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch 2/5  
  Verifying      : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 3/5  
  Verifying      : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64 4/5  
  Verifying      : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64 5/5  
  
Installed:  
mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64 mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch  
mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64  
perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64  
  
Complete!  
[ec2-user@ip-172-31-6-40 ~]$ mysql -h myapppdb.cypfi5rmsrdv.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p  
Enter password:  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MySQL connection id is 33  
Server version: 8.0.46 Source distribution  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
MySQL [(none)]> |
```

Step 6 – Create and Manage Database Objects

- Create a sample database.

```
CREATE DATABASE SampleDB;
```



```
ec2-user@ip-172-31-6-40:~$ sudo yum install mariadb-connector-c mariadb-connector-c-config mariadb105-common mariadb105-libs perl-Sys-Hostname
Installing      : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64      3/5
Installing      : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64        4/5
Installing      : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64           5/5
Running scriptlet: mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64           5/5
Verifying       : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64      1/5
Verifying       : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch 2/5
Verifying       : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64           3/5
Verifying       : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64     4/5
Verifying       : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64        5/5

Installed:
mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64      mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch
mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64          mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Complete!
[ec2-user@ip-172-31-6-40 ~]$ mysql -h myapddb.cypfi5rmsrdv.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 33
Server version: 8.0.46 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE SampleDB;
Query OK, 1 row affected (0.018 sec)

MySQL [(none)]> |
```

- Switch to the database.

USE SampleDB;

```
ec2-user@ip-172-31-6-40:~$ sudo yum install mariadb-connector-c mariadb-connector-c-config mariadb105-common mariadb105-libs perl-Sys-Hostname
Verifying       : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch 2/5
Verifying       : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64           3/5
Verifying       : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64     4/5
Verifying       : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64        5/5

Installed:
mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64      mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch
mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64          mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Complete!
[ec2-user@ip-172-31-6-40 ~]$ mysql -h myapddb.cypfi5rmsrdv.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 33
Server version: 8.0.46 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE SampleDB;
Query OK, 1 row affected (0.018 sec)

MySQL [(none)]> SE SampleDB;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'SE SampleDB' at line 1
MySQL [(none)]> USE SampleDB;
Database changed
MySQL [SampleDB]> |
```

- Create a sample table.

```
CREATE TABLE employees (
  id INT NOT NULL AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  position VARCHAR(100),
  PRIMARY KEY (id)
);
```



```
ec2-user@ip-172-31-6-40:~  
perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64  
  
Complete!  
[ec2-user@ip-172-31-6-40 ~]$ mysql -h myapddb.cypfi5rmsrdv.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p  
Enter password:  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MySQL connection id is 33  
Server version: 8.0.46 Source distribution  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MySQL [(none)]> CREATE DATABASE SampleDB;  
Query OK, 1 row affected (0.018 sec)  
  
MySQL [(none)]> SE SampleDB;  
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version  
for the right syntax to use near 'SE SampleDB' at line 1  
MySQL [(none)]> USE SampleDB;  
Database changed  
MySQL [SampleDB]> CREATE TABLE employees (  
-> id INT NOT NULL AUTO_INCREMENT,  
-> name VARCHAR(100) NOT NULL,  
-> position VARCHAR(100),  
-> PRIMARY KEY (id)  
-> );  
Query OK, 0 rows affected (0.050 sec)  
MySQL [SampleDB]> |
```

- Insert sample records.

```
INSERT INTO employees (name, position)  
VALUES ('John Doe', 'Software Engineer');  
INSERT INTO employees (name, position)  
VALUES ('Jane Smith', 'Project Manager');
```

```
ec2-user@ip-172-31-6-40:~  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MySQL [(none)]> CREATE DATABASE SampleDB;  
Query OK, 1 row affected (0.018 sec)  
  
MySQL [(none)]> SE SampleDB;  
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version  
for the right syntax to use near 'SE SampleDB' at line 1  
MySQL [(none)]> USE SampleDB;  
Database changed  
MySQL [SampleDB]> CREATE TABLE employees (  
-> id INT NOT NULL AUTO_INCREMENT,  
-> name VARCHAR(100) NOT NULL,  
-> position VARCHAR(100),  
-> PRIMARY KEY (id)  
-> );  
Query OK, 0 rows affected (0.050 sec)  
MySQL [SampleDB]> INSERT INTO employees (name, position)  
-> VALUES ('John Doe', 'Software Engineer');  
Query OK, 1 row affected (0.007 sec)  
MySQL [SampleDB]> INSERT INTO employees (name, position)  
-> VALUES ('Jane Smith', 'Project Manager');  
Query OK, 1 row affected (0.006 sec)  
MySQL [SampleDB]> |
```

- Verify the inserted records.

```
SELECT * FROM employees;
```

```
ec2-user@ip-172-31-6-40:~$ mysql
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'SE SampleDB' at line 1
MySQL [(none)]> USE SampleDB;
Database changed
MySQL [SampleDB]> CREATE TABLE employees (
->   id INT NOT NULL AUTO_INCREMENT,
->   name VARCHAR(100) NOT NULL,
->   position VARCHAR(100),
->   PRIMARY KEY (id)
-> );
Query OK, 0 rows affected (0.050 sec)

MySQL [SampleDB]> INSERT INTO employees (name, position)
-> VALUES ('John Doe', 'Software Engineer');
MySQL [SampleDB]> INSERT INTO employees (name, position)
-> VALUES ('Jane Smith', 'Project Manager');
Query OK, 1 row affected (0.007 sec)
Query OK, 1 row affected (0.006 sec)

MySQL [SampleDB]> SELECT * FROM employees;
+----+-----+-----+
| id | name   | position      |
+----+-----+-----+
| 1  | John Doe | Software Engineer |
| 2  | Jane Smith | Project Manager |
+----+-----+-----+
2 rows in set (0.001 sec)

MySQL [SampleDB]> |
```

- Confirm that the records are displayed successfully.

Verification

- The **Amazon RDS MySQL** database was created successfully.
- The database remained private and inaccessible from the public internet.
- The **Amazon EC2** instance connected successfully using the MySQL client.
- A database named **SampleDB** was created.
- The **employees** table was created successfully.
- Sample employee records were inserted and retrieved.
- Communication between **Amazon EC2** and **Amazon RDS** was verified successfully.